

Mikael Elgo
Geophysical Engineering Graduate | Subsurface Geoscience & Geomechanics
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Geophysical Engineering graduate from Institut Teknologi Bandung and Geophysicist at PT Mahakarya Geo Survey with academic and professional experience in seismic processing, reservoir characterization, well-log integration, rock-physics analysis, and quantitative subsurface interpretation. Current academic interests center on rock mechanics, subsurface geomechanics, reservoir geomechanics, wellbore stability, and geoenergy applications. Experienced in integrating seismic and well data, evaluating elastic-property responses, developing reproducible Python workflows, and communicating quantitative subsurface results. Interested in graduate research at the intersection of geophysics, rock mechanics, and subsurface energy engineering.

Research Interests

Rock Mechanics | Subsurface Geomechanics | Reservoir Geomechanics | Wellbore Stability

Education

Bandung Institute of Technology	<i>Bandung, Indonesia</i>
Bachelor of Engineering in Geophysical Engineering, GPA: 3.30/4.0	<i>August 2025</i>
• Honors: Dean's List (4th & 6th semesters for outstanding academic performance) • Relevant Coursework: Reservoir Geophysics, Petrophysics, Structural Geology, Seismic Interpretation, Geostatistics, Well Logging, and Signal Processing.	

Selected Technical Projects

Sandstone Reservoir Characterization Using Rock Physics and Seismic Inversion (Undergraduate Thesis)

Tools: Shearwater Reveal, HRS, Petrel, Interactive Petrophysics

- **Objective:** Characterize thin, interbedded sandstone reservoirs within the Talang Akar Formation by integrating conditioned 3D pre-stack seismic data with well logs from six wells.
- **Methods:** Applied Acoustic Impedance, Shear Impedance, and Extended Elastic Impedance inversion, with familiarity in simultaneous and stochastic inversion concepts; conditioned input data and attenuated multiples using Parabolic Radon Transform in Shearwater Reveal to improve inversion input quality.
- **Output:** Integrated seismic, well-log, and petrophysical interpretation to map elastic-property response and reservoir distribution.
- **Relevance:** Demonstrates rock-physics reasoning, seismic–well integration, inversion QC, and elastic-property interpretation transferable to geomechanical property modeling and quantitative subsurface evaluation.

Self-Directed Geomechanics Study: 1D MEM, Pore Pressure & Wellbore Stability (Ongoing)

Tools: Python, Microsoft Excel, Petrel, Interactive Petrophysics

- **Objective:** Develop a practical foundation in 1D Mechanical Earth Model construction and understand how pore pressure, elastic properties, rock strength, and in-situ stresses influence wellbore stability.
- **Methods:** Followed a structured workflow covering input-log QC, overburden stress from density, sonic-based pore-pressure concepts, dynamic Young's modulus and Poisson's ratio, rock-strength correlations, horizontal stress estimation, calibration, and uncertainty assessment.
- **Output:** Developing Python and Excel calculation exercises using sonic, density, and petrophysical logs to estimate stress gradients, elastic moduli, fracture gradient, and safe mud-weight windows.
- **Relevance:** Connects existing well-log integration and elastic-impedance experience with geomechanical property modeling, drilling-risk evaluation, and quantitative subsurface geomechanics and geoenergy applications.

Self-Directed Study: Blind Well Lithofacies Prediction (FORCE 2020 Well Log Data)

Tools: Python (NumPy, Pandas, scikit-learn, Jupyter), XGBoost, LightGBM, CatBoost, Random Forest, Extra Trees, HistGradientBoosting

- Objective: Predict lithofacies on blind wells from well-log data to test model generalization across unseen wells.
- Methods: Built a reproducible Python pipeline with preprocessing, feature engineering, and group-based well splitting; compared multiple gradient-boosting and tree-ensemble models using accuracy, weighted F1, and macro F1.
- Output: Used petrophysical curves (GR, RHOB, NPHI, DTC, resistivity) to connect classification results to subsurface interpretation context.
- Relevance: Demonstrates multi-well data handling, well-log QC, rigorous validation, model selection, and reproducible Python workflows relevant to digital geoscience, formation evaluation, and data-driven geomechanics.

Professional Experience

Mahakarya Geo Survey

Jakarta, Indonesia

Geophysicist

October 2025 – Present

- Support processing and QC of 2D and 3D marine seismic datasets in ProMAX/SeisSpace, with workflow exposure through Prestack Depth Migration stages.
- Execute data conditioning and processing QC at multiple workflow stages, reviewing parameter choices and inspecting outputs to preserve useful signal and improve overall data quality.
- Perform QC of marine geophysical data (side-scan sonar, sub-bottom profiler, marine magnetometer), checking geometry, navigation, and data integrity, and documenting issues for technical teams.
- Support integration of geophysical results with bathymetric, geotechnical, and metocean datasets for offshore site-investigation reporting, communicating findings in clear technical deliverables.

Pertamina Hulu Rokan

Prabumulih, Indonesia

Geophysicist Intern

April 2025 – August 2025

- Interpreted 3D seismic data and integrated sonic, density, and petrophysical well logs in Petrel and Interactive Petrophysics for the Talang Akar Formation, South Sumatra Basin.
- Evaluated thin, interbedded sandstone-shale reservoirs and applied seismic inversion to assess lateral heterogeneity, depositional variation, and subsurface-property distribution.
- Identified potential bypassed hydrocarbon zones through detailed interpretation and prepared maps, sections, and reservoir-property analyses to support field-development evaluation.
- Presented traceable technical findings to the subsurface team, translating complex seismic and well-log evidence into clear decision-support material.

Lecturer-Led Project (LAPI ITB)

Bandung, Indonesia

Project Assistant

August 2024 – October 2024

- Prepared and validated Seismic Processing Support (SPS) files (shot, receiver, and cross-reference data) for large-scale 3D land seismic projects, supporting acquisition activities for SWATH 38–99 (Antara) and SWATH 1–38 (Balam South).
- Performed geometry quality control and initial seismic processing on 3D seismic datasets using VISTA, covering survey areas of over 4,000 km², to ensure data integrity and accurate subsurface imaging geometry.

Selected Leadership Experience

President of HIMA TG “TERRA” ITB

Bandung, Indonesia | April 2024–March 2025

- Directed the strategy and operations of a geophysical engineering student association with more than 200 members.
- Led and coordinated more than 20 academic, professional, organizational, and social-development programs.
- Collaborated with faculty members, industry professionals, student organizations, and external stakeholders to support program delivery.
- Represented the organization in faculty and inter-university forums and contributed to professional and academic partnerships.

Field Commander of KAT ITB 2023

Bandung, Indonesia | July 2023–August 2023

- Led field coordination and safety implementation for university-wide programs involving more than 10,000 and 2,000 participants.
- Coordinated more than 300 committee members across safety, medical, logistics, security, and operational functions.
- Maintained schedule discipline, participant-flow control, field communication, and operational readiness during high-density activities.
- Enforced HSE procedures and escalation protocols, contributing to the completion of all activities with zero major incidents.
- Responded to operational issues and coordinated timely corrective actions with the relevant teams.

Additional Information

- Languages: Indonesian (native); English (C1 Advanced – EF SET Certificate).
- Academic interest: Rock mechanics, subsurface geomechanics, reservoir geomechanics, wellbore stability, geoenergy, and integrated subsurface characterization.